Week 8 Lab: Insider Threats, Intrusion Detection, and Honeypots

Connect this week's lecture on countermeasures against insider attacks with practical defensive security labs in TryHackMe. You will explore how defenders detect suspicious user and network behaviour through intrusion detection, log analysis, and honeypots. All tasks should be completed inside TryHackMe (AttackBox or your own VM connected to THM).

Learning objectives

The aim of this lab is to help students understand how organisations detect and respond to insider threats and other malicious activity. By completing the activities, students will explain the difference between signature-based and anomaly-based intrusion detection; recognise how audit records and SIEM platforms support investigations; identify suspicious behaviour that may indicate insider misuse or account compromise; and evaluate the role of honeypots in detection and deception.

Requirements

- TryHackMe account and access to AttackBox or your own VM.
- Comfort with basic packet inspection and log analysis.
- A short set of written notes for each activity (screenshots optional).

TryHackMe links

• IDS Fundamentals: https://tryhackme.com/room/idsfundamentals
• Introduction to SIEM: https://tryhackme.com/room/introtosiem
• Log Analysis with SIEM: https://tryhackme.com/room/loganalysiswithsiem
• Introduction To Honeypots: https://tryhackme.com/room/introductiontohoneypots
• Intro to Log Analysis (optional refresher): https://tryhackme.com/room/introtologanalysis

Suggested student workflow

1. Start with IDS Fundamentals to review intrusion detection concepts and the difference between detection and prevention.
2. Use Introduction to SIEM and Log Analysis with SIEM to investigate suspicious events, correlate logs, and identify malicious behaviour.
3. Complete Introduction To Honeypots to understand how decoy systems help detect and study attacker activity.
4. Write short notes explaining how audit logs, behaviour profiling, and honeypots can help detect insider misuse.

Note: Some rooms may require a TryHackMe Premium subscription.